

TMA4300 - Part 1: Stochastic simulation

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2026-02-20

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1 Inversion sampling

Inversion sampling is a method for generating random samples from a target distribution $p(x)$ by using its cumulative distribution function (CDF) $F(x)$. The algorithm works as follows:

1. Compute the CDF $F(x) = \int_{-\infty}^x p(t)dt$.
2. Generate a uniform random number u from the interval $[0, 1]$.
3. Find the value x such that $F(x) = u$. This can be done by computing the inverse CDF, denoted as $F^{-1}(u)$, which gives us the sample from the target distribution.

2 Rejection sampling

Rejection sampling is a method for generating random samples from a target distribution $p(x)$ by using a proposal distribution $q(x)$ that is easy to sample from. The algorithm works as follows:

1. Choose a proposal distribution $q(x)$ and a constant M such that $Mq(x) \geq p(x)$ for all x .
2. Repeat the following steps until you have the desired number of samples:

- Sample x' from the proposal distribution $q(x)$.
- Generate a uniform random number u from the interval $[0, 1]$.
- If $u < \frac{p(x')}{Mq(x')}$, accept x' as a sample from $p(x)$; otherwise, reject it and repeat the process.

Theoretical Acceptance rate

$$\mathbb{E}\left[U < \frac{p(X)}{Mq(X)} \mid X\right] = \int \frac{p(x)}{Mq(x)} q(x) dx = \frac{1}{M} \overbrace{\int p(x) dx}^{=1} = \frac{1}{M}$$

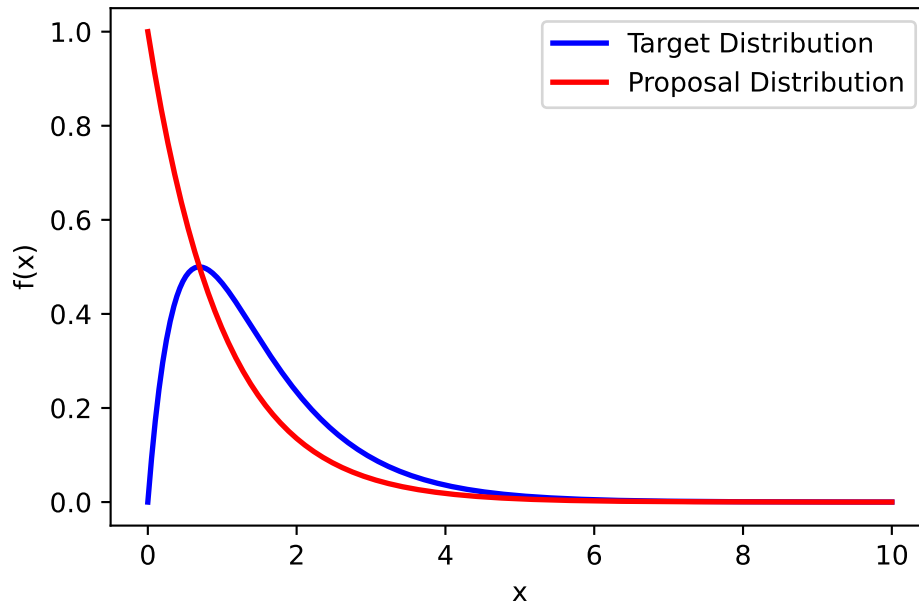
The efficiency of rejection sampling depends on how well the proposal distribution $q(x)$ approximates the target distribution $p(x)$. If M is large, the acceptance rate will be low, leading to many rejected samples. Therefore, it is crucial to choose a proposal distribution that closely matches the shape of the target distribution to improve efficiency.

💡 Example

Suppose you want to sample from

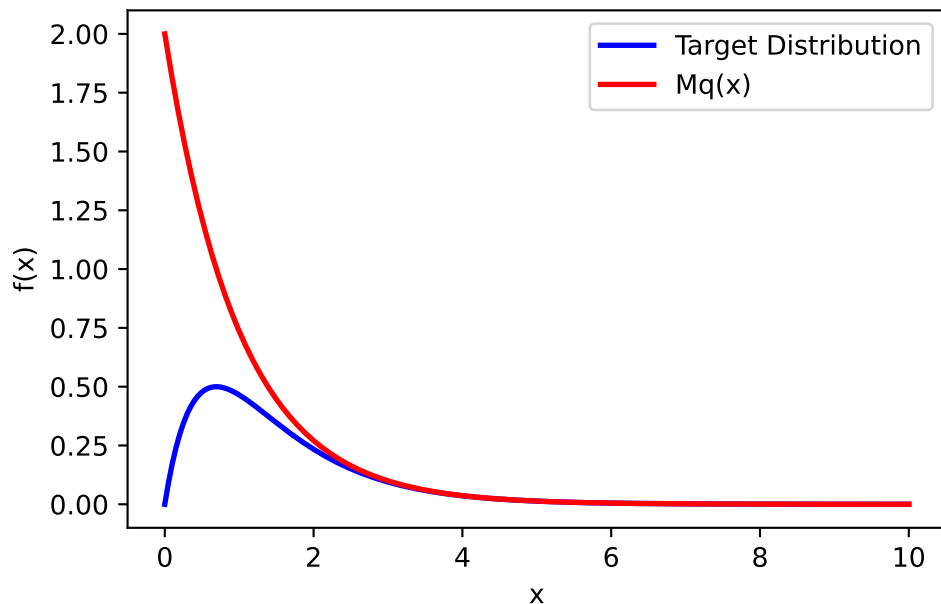
$$p(x) = \begin{cases} 2e^{-x}(1 - e^{-x}) & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

having $q(x) = e^{-x}$ for $x \geq 0$ as proposal distribution.



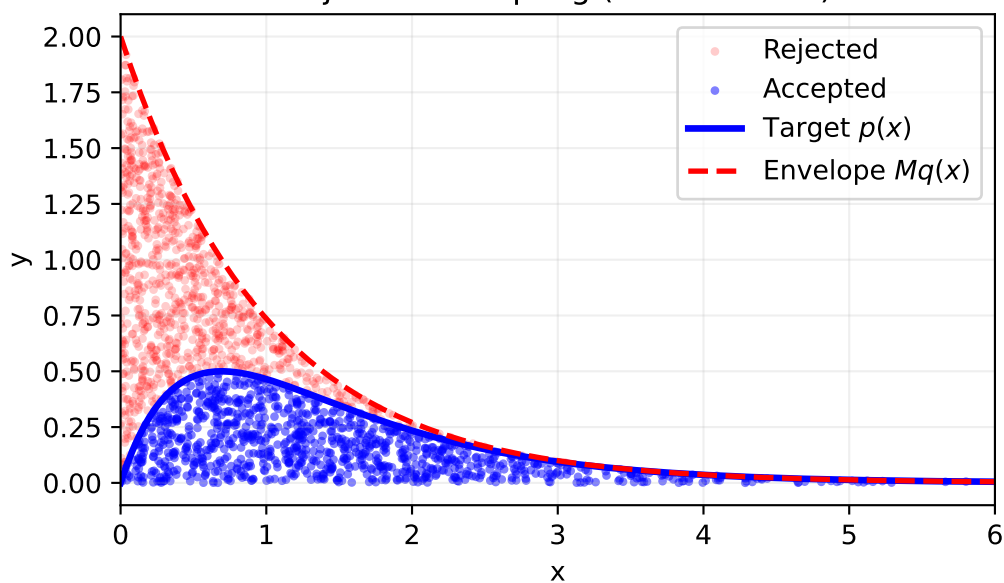
We have to choose a constant M such that $Mq(x) \geq p(x)$ for all x :

$$M = \max_{x \geq 0} \frac{p(x)}{q(x)} = \max_{x \geq 0} \frac{2e^{-x}(1 - e^{-x})}{e^{-x}} = \max_{x \geq 0} 2(1 - e^{-x}) = 2$$



We can now try to run point 2b - 2c until we reach 1000 accepted samples:

Rejection Sampling (Rate: 0.5277)



We can see that the empirical acceptance rate is close to the theoretical one, given by

$$\text{Acceptance Rate} = \frac{1}{M} = \frac{1}{2} = 0.5$$

3 Convergence of random variables

Let $\{X_n\}_{n=1}^{\infty}$ be a sequence of random variables.

i Almost sure convergence

If

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} X_n = X\right) = 1$$

i.e.

$$\mathbb{P}\left(\left\{\omega \in \Omega : \lim_{n \rightarrow \infty} X_n(\omega) = X\right\}\right) = 1$$

Then

$$X_n \xrightarrow{\text{a.s.}} X \quad \text{as } n \rightarrow \infty$$

i Mean square convergence

If

$$\lim_{n \rightarrow \infty} \mathbb{E}\left((X_n - X)^2\right) = 0$$

Then

$$X_n \xrightarrow{L_2} X$$

i Convergence in probability

If

$$\lim_{n \rightarrow \infty} \mathbb{P}\left(|X_n - X| < \varepsilon\right) = 1$$

Then

$$X_n \xrightarrow{p} X$$

i Convergence implications

- a.e. convergence $\implies$ probability convergence
- L^p convergence $\implies$ probability convergence
- probability convergence $\implies$ convergence in distribution

Strong Law of Large Numbers (SLLN)

Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ where $X_1, X_2, \dots$ are iid with finite $\mathbb{E}(X_i) = \mu$.
Then:

$$\bar{X}_n \xrightarrow{a.s.} \mu$$

Weak LLN:

$$\bar{X}_n \xrightarrow{p} \mu$$

4 Monte Carlo Integration

Aim: Estimate $\mathbb{E}[h(\mathbf{X})]$ where $\mathbf{X} \sim f(x)$.

Example of possible estimates

Monte Carlo is not only used to estimate the mean of the random variable!

- Choosing $h(X) = X$, we can estimate the mean $\mathbb{E}[X]$
- Choosing $h(X) = (X - \mu)^2$, we can estimate the variance $\text{Var}(X) = \mathbb{E}[(X - \mu)^2]$
- Choosing $h(X) = I(X > a)$, we can estimate the tail probability $\mathbb{P}(X > a)$

Procedure: Simulate $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} f(x)$ and compute the empirical average:

$$\hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n h(X_i)$$

Properties of the Estimator

- **Unbiasedness:**

$$\mathbb{E}[\hat{\theta}_n] = \mathbb{E}\left(\frac{1}{n} \sum_{i=1}^n h(X_i)\right) = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[h(X_i)] = \mathbb{E}[h(X)]$$

- **Convergence (Strong Law of Large Numbers):** As $n \rightarrow \infty$, the estimator converges almost surely to the true value:

$$\frac{1}{n} \sum_{i=1}^n h(X_i) \xrightarrow{a.s.} \mathbb{E}[h(X)]$$

- **Error & Rate of Convergence (Central Limit Theorem):** Assuming $\sigma_h^2 = \text{Var}(h(X)) < \infty$, the estimation error is asymptotically normally distributed:

$$\sqrt{n} (\hat{\theta}_n - \mathbb{E}[h(X)]) \xrightarrow{d} \mathcal{N}(0, \sigma_h^2)$$

Key Takeaway: This implies the standard error decreases at a rate of $\mathcal{O}(1/\sqrt{n})$, regardless of the dimensionality of X . This is why Monte Carlo shines in high dimensions!